第四讲

简单截面数据
的经济计量方法
（三）

总结：高斯-马尔科夫定理

- 在假定1-假定4的条件下，参数的OLS估计是最优线性无偏估计(BLUE:Best Linear unbiased setimator)
  - 没有其他的估计会比OLS估计更好！！
- 问：社会经济问题研究中，高斯-马尔科夫假定哪些不易满足？
  - 假定：$\mathrm{Var}(u|x) = \sigma^2$.不易满足(异方差)，$u$的方差与$\boldsymbol{x}$取值相关

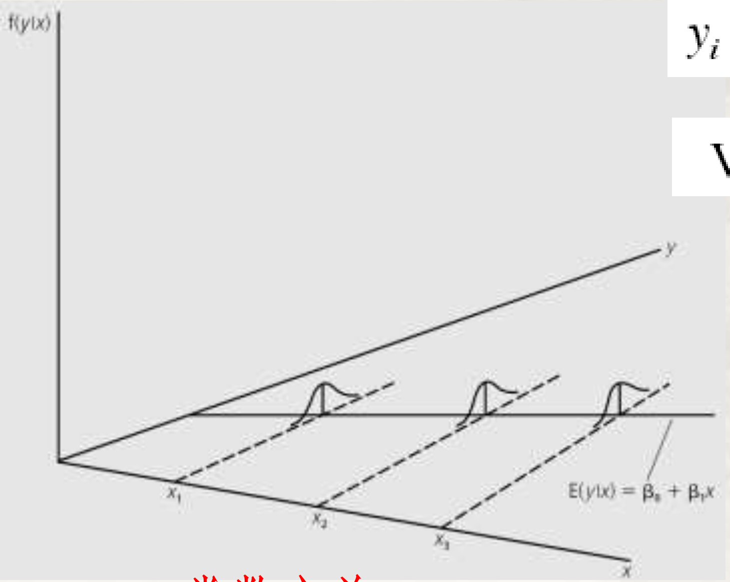
- 不满足假定4情况下：异方差
  - 异方差导致的问题
  - 异方差的诊断
  - 异方差的处理策略

- 高斯-马尔科夫假定4：同方差性 $\mathrm{Var}(u|x) = \sigma^2$.

$$y = \beta_0 + \beta_1 x + u,$$

⬇

$\mathrm{Var}(u|x) = \mathrm{Var}(y|x)$ $\mathrm{Var}(y|x) = \sigma^2.$



$$y_i = \beta_0 + \beta_1 x_i + u_i.$$

$$\mathrm{Var}(u_i|x_i) = \sigma_i^2,$$

常数方差

- 同方差假定对估计OLS估计量方差的意义：
  - 一元模型中：

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^{n}(x_i - \bar{x})u_i}{s_x^2} = \beta_1 + (1/s_x^2)\sum_{i=1}^{n} d_i u_i,$$

$$\begin{aligned}\mathrm{Var}(\hat{\beta}_1) &= (1/s_x^2)^2 \mathrm{Var}\left(\sum_{i=1}^{n} d_i u_i\right) = (1/s_x^2)^2\left(\sum_{i=1}^{n} d_i^2 \mathrm{Var}(u_i)\right) \\ &= (1/s_x^2)^2\left(\sum_{i=1}^{n} d_i^2\sigma^2\right) \quad [\text{since } \mathrm{Var}(u_i) = \sigma^2 \text{ for all } i] \\ &= \sigma^2(1/s_x^2)^2\left(\sum_{i=1}^{n} d_i^2\right) = \sigma^2(1/s_x^2)^2 s_x^2 = \sigma^2/s_x^2,\end{aligned}$$

  - 多元模型中：

$$\mathrm{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\mathrm{SST}_j(1 - R_j^2)}$$ $$\mathrm{SST}_j = \sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2$$

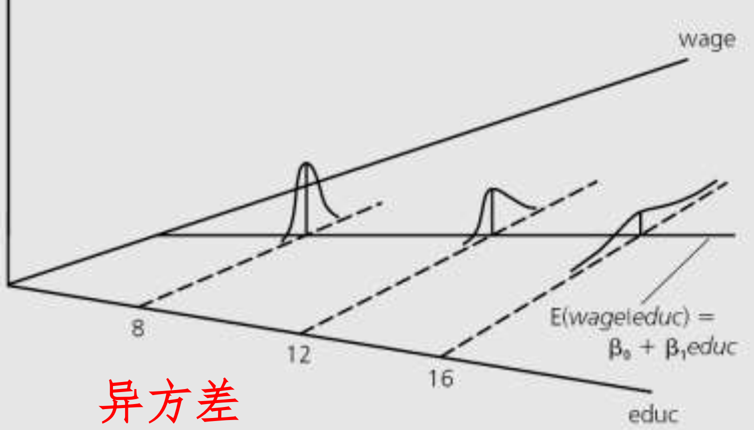
- 异方差将导致前述估计OLS估计量方差的方法失效
  - 以一元模型为例：

$$y_i = \beta_0 + \beta_1 x_i + u_i. \quad \mathrm{Var}(u_i|x_i) = \sigma_i^2,$$

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^{n}(x_i - \bar{x})u_i}{s_x^2} = \beta_1 + (1/s_x^2)\sum_{i=1}^{n} d_i u_i.$$

$$\begin{aligned}
\mathrm{Var}(\hat{\beta}_1) &= (1/s_x^2)^2 \mathrm{Var}\left(\sum_{i=1}^{n} d_i u_i\right) = (1/s_x^2)^2\left(\sum_{i=1}^{n} d_i^2 \mathrm{Var}(u_i)\right) \\
&\cancel{=} \boxed{(1/s_x^2)^2\left(\sum_{i=1}^{n} d_i^2 \sigma^2\right)} \qquad [\text{since } \mathrm{Var}(u_i) = \sigma^2 \text{ for all } i] \\
&= \sigma^2 (1/s_x^2)^2\left(\sum_{i=1}^{n} d_i^2\right) = \sigma^2 (1/s_x^2)^2 s_x^2 = \sigma^2/s_x^2,
\end{aligned}$$

再例：收入与消费



- 异方差下：

$$\mathrm{Var}(\hat{\beta}_1) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sigma_i^2}{\mathrm{SST}_x^2}$$

下标i表示误差方差与x的第i水平有关

异方差的处理策略(一):稳健统计量

- 不放弃OLS估计，且采用异方差稳健统计量检验
  - 异方差稳健统计量（大样本下使用）（第8章）

$$t = \frac{\text{estimate} - \text{hypothesized value}}{\text{standard error}}$$

异方差稳健标准误

- 异方差稳健统计量在大样本下近似服从t分布

异方差的处理策略(一):稳健统计量

- 异方差稳健标准误的常见估计方法：怀特法
  - 一元模型中 $\text{Var}(\hat{\beta}_1)$:

$$\text{Var}(\hat{\beta}_1) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2\sigma_i^2}{\text{SST}_x^2} \Rightarrow \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2\hat{u}_i^2}{\text{SST}_x^2}$$

用y对x回归得到的OLS残差

    - 可证明$n\times\text{Var}(\hat{\beta}_1)$依概率收敛于$\frac{\sum_{i=1}^{n}(x_i-\bar{x})^2\sigma_i^2}{\sigma_x^2}$

  - 多元模型中：

$$\hat{\text{Var}}(\hat{\beta}_j) = \frac{\sum_{i=1}^{n}\hat{r}_{ij}^2\hat{u}_i^2}{\text{SSR}_j^2},$$

为x_j对其他解释变量回归(辅助回归)的第i个残差；分母为辅助回归(x_j为被解释变量)的残差平方和

■ 例：教育回报率问题

```
> summary(lgWAGE_lm1<-lm(log(WAGE)~EDUC+EXPER,data=Data))

Call:
lm(formula = log(WAGE) ~ EDUC + EXPER, data = Data)

Residuals:
     Min       1Q   Median       3Q      Max
-2.05800 -0.30136 -0.04539  0.30601  1.44425

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.216854   0.108595   1.997   0.0464 *
EDUC        0.097936   0.007622  12.848  < 2e-16 ***
EXPER       0.010347   0.001555   6.653 7.24e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4614 on 523 degrees of freedom
Multiple R-squared:  0.2493,    Adjusted R-squared:  0.2465
F-statistic: 86.86 on 2 and 523 DF,  p-value: < 2.2e-16

> sqrt(diag(vcov(lgWAGE_lm1)))
(Intercept)        EDUC       EXPER
0.108595022 0.007622398 0.001555139
```

vcov函数:得到估计量的协方差阵

```
. reg LNWAGE EDU EXPER

      Source |       SS       df       MS              Number of obs =     526
-------------+------------------------------           F(2, 523)     =   86.86
       Model |  36.9850401     2  18.4925201           Prob > F      =  0.0000
    Residual |   111.34471   523  .212896195           R-squared     =  0.2493
-------------+------------------------------           Adj R-squared =  0.2465
       Total |   148.32975   525  .282532857           Root MSE      =  .46141

------------------------------------------------------------------------------
      LNWAGE |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        EDUC |   .0979356   .0076224    12.85   0.000     .0829613    .1129099
       EXPER |   .0103469   .0015551     6.65   0.000     .0072919     .013402
       _cons |   .2168544    .108595     2.00   0.046     .0035183    .4301904
------------------------------------------------------------------------------

. vce

Covariance matrix of coefficients of regress model

        e(V) |       EDUC       EXPER       _cons
-------------+------------------------------------
        EDUC |   .0000581
       EXPER |  3.551e-06   2.418e-06
       _cons | -.00079033  -.00008576   .01179288
```

显示估计量的协方差阵

- 例：对教育回报率问题，采用怀特法(HC0)估计稳健标准误：

library("sandwich")

```
> sqrt(diag(vcovHC(lgWAGE_lm1,type="HC0")))
(Intercept)        EDUC       EXPER
0.114183256 0.008072781 0.001600762
```

vcovHC实现：type= “const” 为同方差假定下的普通标准误

- 检验时指定估计标准误

library("lmtest")

```
> coeftest(lgWAGE_lm1,vcov=vcovHC(lgWAGE_lm1,type="HC0"))

t test of coefficients:

             Estimate Std. Error t value  Pr(>|t|)
(Intercept) 0.2168544  0.1141833  1.8992   0.05809 .
EDUC        0.0979356  0.0080728 12.1316 < 2.2e-16 ***
EXPER       0.0103469  0.0016008  6.4638 2.343e-10 ***
```

```
. reg LNWAGE EDU EXPER, vce(r)

Linear regression                               Number of obs  =       526
                                                F(2, 523)      =     74.64
                                                Prob > F       =    0.0000
                                                R-squared      =    0.2493
                                                Root MSE       =    .46141

------------------------------------------------------------------------------
             |               Robust
      LNWAGE |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        EDUC |   .0979356   .0080959    12.10   0.000     .0820311    .1138401
       EXPER |   .0103469   .0016053     6.45   0.000     .0071932    .0135007
       _cons |   .2168544   .1145103     1.89   0.059    -.0081022     .441811
------------------------------------------------------------------------------
```

$$\mathrm{HC1}:\quad \omega_i = \frac{n}{n-k}\hat{\varepsilon}_i^2$$

针对小样本的改进策略

- 还有多种估计，通常均大于同方差假设下的标准误

```
> t(sapply(c("const","HC0","HC1","HC2","HC3","HC4"),
+     FUN=function(x) sqrt(diag(vcovHC(lgWAGE_lm1,type=x)))))
      (Intercept)        EDUC       EXPER
const   0.1085950 0.007622398 0.001555139
HC0     0.1141833 0.008072781 0.001600762
HC1     0.1145103 0.008095901 0.001605347
HC2     0.1150595 0.008137424 0.001607357
HC3     0.1159543 0.008203438 0.001613998
HC4     0.1169298 0.008277719 0.001612426
```

- 无论是否存在异方均可采用异方差稳健t统计量（要求大样本）
- 因各种估计均大于同方差假设下的标准误，同方差假定成立下最好不使用稳健统计量
- 对是否存在异方差进行诊断是必要的！

异方差的诊断

- 方法一：布罗施帕甘检验(Breusch-Pagan test , BP test)（第8章）
  - 原假设：同方差

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + u,$$

$$\mathrm{H}_0: \mathrm{Var}(u|x_1, x_2, \ldots, x_k) = \sigma^2.$$

    - 因为：$\mathrm{Var}(u|\boldsymbol{x}) = \mathrm{E}(u^2|\boldsymbol{x})$
    - 原假设为：$\mathrm{H}_0: \mathrm{E}(u^2|x_1, x_2, \ldots, x_k) = \mathrm{E}(u^2) = \sigma^2.$
  - 思路：检验u^2是否与一个或多个解释变量相关；若u^2是$\boldsymbol{x}$的某种函数(线性或非线性)，拒绝原假设

$$u^2 = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \ldots + \delta_k x_k + v,$$

给定X下均值为0的误差项

$$\mathrm{H}_0: \delta_1 = \delta_2 = \ldots = \delta_k = 0.$$

异方差检验以模型设置正确为前提!!

问：如何得到u?

- BP检验的基本步骤：
  - 用OLS估计模型： $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + u,$
  - 得到OLS残差平方： $\hat{u}^2$
  - 用OLS估计模型： $\hat{u}^2 = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \ldots + \delta_k x_k + error$ 其R^2记为：$R^2_{\hat{u}^2}$.
    - 对估计量进行联合显著性检验：
      - F统计量： $F = \frac{R^2_{\hat{u}^2}/k}{(1 - R^2_{\hat{u}^2})/(n - k - 1)}, \sim F_{k,n-k-1}$ $\frac{R^2/k}{(1 - R^2)/(n - k - 1)}$
      - LM统计量：

$$LM = n \cdot R^2_{\hat{u}^2}. \sim \chi^2_k$$

LM形式的检验称为布罗施帕甘检验（自由度为k）

- BP检验的另一种形式

$$\hat{u}^2 = \delta_0 + \delta_1 \hat{y} + error,$$

BP检验示例：教育回报率问题

```
> summary(lm(lgWAGE_lm1$residuals^2~EDUC+EXPER,data=Data))

Call:
lm(formula = lgWAGE_lm1$residuals^2 ~ EDUC + EXPER, data = Data)

Residuals:
    Min      1Q  Median      3Q     Max
-0.2876 -0.1711 -0.0978  0.0444  4.0773

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.049323   0.079342   0.622  0.53445
EDUC        0.008186   0.005569   1.470  0.14219
EXPER       0.003498   0.001136   3.078  0.00219 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3371 on 523 degrees of freedom
Multiple R-squared:  0.0184,    Adjusted R-squared:  0.01465
F-statistic: 4.903 on 2 and 523 DF,  p-value: 0.007762

> (LM<-0.0184*526)
[1] 9.6784
> 1-pchisq(q=LM,df=2)   #计算概率P值
[1] 0.007913382
```

拒绝原假设，存在异方差

$$H_0: \delta_1 = \delta_2 = \ldots = \delta_k = 0.$$

$$\hat{u}^2 = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \ldots + \delta_k x_k + error$$

rhs:BP检验采用模型右项

iid:残差只需满足独立同分布(无需正态分布)

```
. estat hettest,rhs iid

Breusch-Pagan / Cook-Weisberg test for heteroskedasticity
         Ho: Constant variance
         Variables: EDUC EXPER

         chi2(2)      =     9.68
         Prob > chi2  =   0.0079
```

$$\hat{u}^2 = \delta_0 + \delta_1 \hat{y} + error,$$

```
. estat hettest,iid

Breusch-Pagan / Cook-Weisberg test for heteroskedasticity
         Ho: Constant variance
         Variables: fitted values of LNWAGE

         chi2(1)      =     4.15
         Prob > chi2  =   0.0417
```

```
> bptest(lgWAGE_lm1)   #BP检验

        studentized Breusch-Pagan test

data:  lgWAGE_lm1
BP = 9.681, df = 2, p-value = 0.007903
```

·调整的BP检验：基于学生化残差的平方

异方差检验以模型设置正确为前提!!

异方差的诊断

- 方法二：怀特检验(White test for heteroskedasticity)
  - 思路：类似布罗施帕甘检验，并认为若同方差则u^2与所有解释变量、所有解释变量的平方、解释变量的交乘项都不相关

$$\hat{u}^2 = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \delta_3 x_3 + \delta_4 x_1^2 + \delta_5 x_2^2 + \delta_6 x_3^2 + \delta_7 x_1 x_2 + \delta_8 x_1 x_3 + \delta_9 x_2 x_3 + error.$$

  - 若同方差，则上述模型的所有系数同时为零

- 异方差性的怀特检验基本步骤：
  - 用OLS估计模型： $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + u,$
    - 得到OLS残差平方和拟合值及平方： $\hat{u}^2$
  - 用OLS估计模型：

$$\hat{u}^2 = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \delta_3 x_3 + \delta_4 x_1^2 + \delta_5 x_2^2 + \delta_6 x_3^2 + \delta_7 x_1 x_2 + \delta_8 x_1 x_3 + \delta_9 x_2 x_3 + error.$$

其R^2记为： $R^2_{\hat{u}^2}.$

    - 对估计量进行联合显著性检验：
      - F统计量

$$F = \frac{R^2_{\hat{u}^2}/k}{(1 - R^2_{\hat{u}^2})/(n - k - 1)}, \sim F_{k,n-k-1}$$

(k为全部变元个数)

      - LM统计量(BP)

$$LM = n \cdot R^2_{\hat{u}^2}. \sim \chi^2_k$$

(k为全部变元个数)

- 怀特检验的改进

$$\hat{u}^2 = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \delta_3 x_3 + \delta_4 x_1^2 + \delta_5 x_2^2 + \delta_6 x_3^2 + \delta_7 x_1 x_2 + \delta_8 x_1 x_3 + \delta_9 x_2 x_3 + error.$$

  - 因所估计参数较多，改进为：引入拟和值及其平方（是解释变量平方、所有交乘项的一个特殊函数）
- 步骤：
  - 用OLS估计模型： $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + u,$
    - 得到OLS残差平方和拟合值及平方： $\hat{u}^2$ $\hat{y}$
  - 用OLS估计模型： $\hat{u}^2 = \delta_0 + \delta_1 \hat{y} + \delta_2 \hat{y}^2 + error,$
    - 对估计量进行联合显著性检验
      - F统计量： $F = \dfrac{R_{\hat{u}^2}^2/k}{(1 - R_{\hat{u}^2}^2)/(n - k - 1)}, \sim F_{k,n-k-1}$ (k=2)
      - LM统计量： $LM = n \cdot R_{\hat{u}^2}^2. \sim \chi_k^2$ (k=2)

- R的异方差性的怀特检验及改进的怀特检验示例：
  - 例：教育回报率问题

$$\hat{u}^2 = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \delta_3 x_3 + \delta_4 x_1^2 + \delta_5 x_2^2 + \delta_6 x_3^2 + \delta_7 x_1 x_2 + \delta_8 x_1 x_3 + \delta_9 x_2 x_3 + error.$$

```
> bptest(lgWAGE_lm1,~EDUC*EXPER+I(EDUC^2)+I(EXPER^2),data=Data)

        studentized Breusch-Pagan test

data:  lgWAGE_lm1
BP = 18.2971, df = 5, p-value = 0.002596
```

公式表示解释变量包含：
自身、平方项和交乘项

```
. estat imtest,white

White's test for Ho: homoskedasticity
         against Ha: unrestricted heteroskedasticity

         chi2(5)      =     18.30
         Prob > chi2  =    0.0026
```

```
> bptest(lgWAGE_lm1,~lgWAGE_lm1$fitted.values+I(lgWAGE_lm1$fitted.values^2))

        studentized Breusch-Pagan test

data:  lgWAGE_lm1
BP = 7.7455, df = 2, p-value = 0.0208
```

$$\hat{u}^2 = \delta_0 + \delta_1 \hat{y} + \delta_2 \hat{y}^2 + error,$$

- 引入所有二次交乘项，拒绝原假设，存在异方差性
- 利用拟合值检验，拒绝原假设，存在异方差性

异方差的处理策略（二）：加权最小二乘

- 加权最小二乘估计(weighted least squares, WLS)
  - 存在异方差，且已知：

    $$\mathrm{Var}(u|\boldsymbol{x}) = \sigma^2 h(\boldsymbol{x}),$$

    - $h(x)$大于0，是解释变量的某种函数
    - 例：

      $$sav_i = \beta_0 + \beta_1 inc_i + u_i$$

      $$\mathrm{Var}(u_i|inc_i) = \sigma^2 inc_i.$$

      误差项的方差同收入成正比

      - 收入越高，储蓄的方差越大

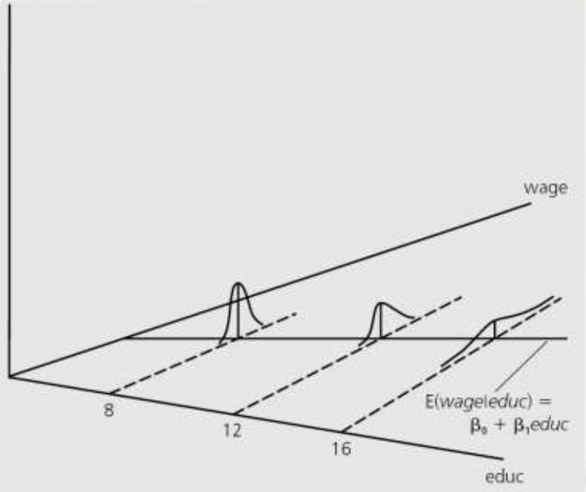
$$\sigma_i^2 = \mathrm{Var}(u_i|\boldsymbol{x}_i) = \sigma^2 h(\boldsymbol{x}_i) = \sigma^2 h_i$$

- 加权最小二乘估计（WLS）
  - OLS中每个观测的残差在平方和中有相同的权重

$$\min_{b_0,b_1} \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2,$$

  - WLS：由给定x下误差项的方差决定观测的残差在平方和中的权重

给定x下误差项的方差越大，权重越小。反之



- 即求下式最小下的参数估计：

$$\sum_{i=1}^{n} (y_i - b_0 - b_1 x_{i1} - b_2 x_{i2} - \ldots - b_k x_{ik})^2 / h_i$$

$$\sigma_i^2 = \mathrm{Var}(u_i|\boldsymbol{x}_i) = \sigma^2 h(\boldsymbol{x}_i) = \sigma^2 h_i$$

将 $1/\sqrt{h_i}$ 代入

$$\sum_{i=1}^{n} (y_i^* - b_0 x_{i0}^* - b_1 x_{i1}^* - b_2 x_{i2}^* - \ldots - b_k x_{ik}^*)^2.$$

1/h_i为权重，h_i越大，权重越小

- 加权最小二乘估计（WLS）
  - 对变换后的如下数据再进行OLS估计：

$$y_i^* = \beta_0 x_{i0}^* + \beta_1 x_{i1}^* + \dots + \beta_k x_{ik}^* + u_i^*,$$

$$\Updownarrow$$

$$y_i/\sqrt{h_i} = \beta_0/\sqrt{h_i} + \beta_1(x_{i1}/\sqrt{h_i}) + \beta_2(x_{i2}/\sqrt{h_i}) + \dots + \beta_k(x_{ik}/\sqrt{h_i}) + (u_i/\sqrt{h_i})$$

该方程满足同方差假定

误差项的方差：$\mathrm{E}\left((u_i/\sqrt{h_i})^2\right) = \mathrm{E}(u_i^2)/h_i = (\sigma^2 h_i)/h_i = \sigma^2.$

  - 满足经典线性模型假定，OLS估计为BLUE
  - 估计量称为原方程的加权最小二乘估计量，是广义最小二乘(Generalized least squares, GLS)估计量的特例

异方差的处理策略(二)：加权最小二乘

- 加权最小二乘估计（WLS）
  - 问题：如何得到h_i
    - 前例中,$h_i = h(x_i)$是单个解释变量x的某种函数，在多元中具有不合理性
    - $h(\boldsymbol{x})$的具体形式：通常是关于解释变量全体的一个非负函数

- 加权最小二乘估计中如何确定h_i？
  - 可借助一个假定函数估计h_i的取值，记为$\hat{h}_i$
  - 由此得到的估计称为可行的GLS（Feasible GLS,FGLS）估计
  - 假定：

$$h(\boldsymbol{x}) = \exp(\delta_0 + \delta_1 x_1 + \delta_2 x_2 + \ldots + \delta_k x_k).$$

$$\mathrm{Var}(u|\boldsymbol{x}) = \sigma^2 \exp(\delta_0 + \delta_1 x_1 + \delta_2 x_2 + \ldots + \delta_k x_k),$$

    - 选择指数函数的原因是确保h_i为正
  - 于是：

$\mathrm{Var}(u|\boldsymbol{x}) = \mathrm{E}(u^2|\boldsymbol{x})$ $u^2 = \sigma^2 \exp(\delta_0 + \delta_1 x_1 + \delta_2 x_2 + \ldots + \delta_k x_k) v,$

v=1，保证下式中e的期望为0

    - 估计出参数，便可得到h_i的估计

$$\log(u^2) = \alpha_0 + \delta_1 x_1 + \delta_2 x_2 + \ldots + \delta_k x_k + e,$$

加权最小二乘估计中的h_i

- 估计h_i的基本步骤： $y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + u$
  - 做y对$x_1, x_2, \ldots, x_k$的回归，得到残差 $\hat{u}$.
  - 计算残差平方，并取对数得到 $\log(\hat{u}^2)$
  - 估计方程：

$$\log(u^2) = \alpha_0 + \delta_1 x_1 + \delta_2 x_2 + \ldots + \delta_k x_k + e,$$

  - 计算拟合值：$\hat{g}$.
  - 得到h_i的估计值： $\hat{h}_i = \exp(\hat{g}_i)$.

- 估计h_i后，以 $1/\hat{h}_i$ 为权数，采用WLS估计参数，得到原方程的FGLS估计量

FGLS示例：教育回报率问题

```
> auxreg<-lm(log(lgWAGE_lm1$residuals^2)~EDUC+EXPER,data=Data)  #辅助回归
> summary(lgWAGE_fgls<-lm(log(WAGE)~EDUC+EXPER,
+          weights=1/exp(fitted(auxreg)),data=Data))  #FGLS可行的加权最小二乘

Call:
lm(formula = log(WAGE) ~ EDUC + EXPER, data = Data, weights = 1/exp(fitted(auxreg)))

Weighted Residuals:
    Min      1Q  Median      3Q     Max 
-9.7302 -1.2825 -0.1278  1.1727  6.1649 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.252805   0.099667   2.536   0.0115 *  
EDUC        0.091670   0.007275  12.601  < 2e-16 ***
EXPER       0.013106   0.001656   7.912 1.52e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.

Residual standard error: 1.794 on 523
Multiple R-squared:  0.2562,    Adjus
F-statistic: 90.06 on 2 and 523 DF,
```

FGLS估计

```
> summary(lgWAGE_lm1<-lm(log(WAGE)~EDUC+EXPER,data=Data))

Call:
lm(formula = log(WAGE) ~ EDUC + EXPER, data = Data)

Residuals:
     Min       1Q   Median       3Q      Max 
-2.05800 -0.30136 -0.04539  0.30601  1.44425 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.216854   0.108595   1.997   0.0464 *  
EDUC        0.097936   0.007622  12.848  < 2e-16 ***
EXPER       0.010347   0.001555   6.653 7.24e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

OLS估计

FGLS示例：教育回报率问题

```
. quietly reg LNWAGE EDU EXPER

. predict e1,residual

. generate e2=e1^2

. generate lne2=log(e2)

. regress lne2 EDU EXPER
```

Source	SS	df	MS
Model	63.4471621	2	31.723581
Residual	2168.24598	523	4.14578581
Total	2231.69314	525	4.25084408

Number of obs	=	526
F(2, 523)	=	7.65
Prob > F	=	0.0005
R-squared	=	0.0284
Adj R-squared	=	0.0247
Root MSE	=	2.0361

lne2	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
EDUC	.0732226	.0336365	2.18	0.030	.0071433	.1393019
EXPER	.0257569	.0068626	3.75	0.000	.0122753	.0392386
_cons	-4.093258	.4792139	-8.54	0.000	-5.034679	-3.151838

```
. predict lne2f
(option xb assumed; fitted values)

. generate e2f=exp(lne2f)

. reg LNWAGE EDU EXPER [aw=1/e2f]
(sum of wgt is 8,588.23265805587)
```

Source	SS	df	MS
Model	35.5151985	2	17.7575992
Residual	103.121641	523	.19717331
Total	138.63684	525	.264070171

Number of obs	=	526
F(2, 523)	=	90.06
Prob > F	=	0.0000
R-squared	=	0.2562
Adj R-squared	=	0.2533
Root MSE	=	.44404

LNWAGE	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
EDUC	.0916701	.0072747	12.60	0.000	.077379	.1059612
EXPER	.0131062	.0016565	7.91	0.000	.009852	.0163604
_cons	.2528054	.0996671	2.54	0.011	.0570083	.4486024

问：以上处理是否确实消除了异方差性？

- 对变换后的教育回报率模型进行异方差检验(改进怀特检验)

$$\hat{u}^2 = \delta_0 + \delta_1\hat{y} + \delta_2\hat{y}^2 + error,$$

```
> bptest(lgWAGE_fgls,~lgWAGE_fgls$fitted.values+I(lgWAGE_fgls$fitted.values^2))

        studentized Breusch-Pagan test

data:  lgWAGE_fgls
BP = 10.1308, df = 2, p-value = 0.006312
```

拒绝异方差检验的原假设，仍存在异方差！

- 若没有完全消除异方差，一般处理方法：
  - 采用稳健的FGLS标准误
    - 以一元为例：采用FGLS估计，且基于稳健标准误进行统计检验

$$\frac{\sum_{i=1}^{n}(x_i-\bar{x})^2\hat{u}_i^2}{\mathrm{SST}_x^2} \leftarrow \left(\frac{\hat{u}_i}{\sqrt{\hat{h}_i}}\right)^2$$

$$y_i^* = \beta_0 x_{i0}^* + \beta_1 x_{i1}^* + \ldots + \beta_k x_{ik}^* + u_i^*,$$

$$\Updownarrow$$

$$y_i/\sqrt{h_i} = \beta_0/\sqrt{h_i} + \beta_1(x_{i1}/\sqrt{h_i}) + \beta_2(x_{i2}/\sqrt{h_i}) + \ldots + \beta_k(x_{ik}/\sqrt{h_i}) + (u_i/\sqrt{h_i})$$

FGLS示例：教育回报率问题，采用稳健的WLS标准误

```
> coeftest(lgWAGE_fgls,vcov=vcovHC(lgWAGE_fgls,type="HC0"))

t test of coefficients:

             Estimate Std. Error t value  Pr(>|t|)
(Intercept) 0.2528054  0.1250019  2.0224   0.04364 *
EDUC        0.0916701  0.0090254 10.1569 < 2.2e-16 ***
EXPER       0.0131062  0.0016002  8.1901 2.013e-15 ***
```

```
. reg LNWAGE EDU EXPER [aw=1/e2f], vce(r)
(sum of wgt is 8,588.23265805587)

Linear regression                               Number of obs     =        526
                                                F(2, 523)         =      62.56
                                                Prob > F          =     0.0000
                                                R-squared         =     0.2562
                                                Root MSE          =     .44404
```

$$\text{HC1}: \quad \omega_i = \frac{n}{n-k}\hat{\varepsilon}_i^2$$

LNWAGE	Coef.	Robust Std. Err.	t	P>|t|	[95% Conf.	Interval]
EDUC	.0916701	.0090512	10.13	0.000	.0738889	.1094513
EXPER	.0131062	.0016048	8.17	0.000	.0099535	.0162589
_cons	.2528054	.1253599	2.02	0.044	.0065346	.4990762

- **总结：**
  - 若原始模型等方差性假设不成立且$h(\boldsymbol{x})$未知，应采用FGLS，且一般采用稳健的FGLS标准误
- 问：为什么变换后的模型仍存在异方差？答：$h(x)$函数形式可能不正确！
  - h_i是FGLS的关键

$$\text{var}\left(\frac{u_i}{\sqrt{\hat{h}_i}}\right) = \frac{\text{var}(u_i)}{\hat{h}_i} = \frac{\sigma^2 h_i}{\hat{h}_i} \Rightarrow \sigma^2$$

$$y_i/\sqrt{h_i} = \beta_0/\sqrt{h_i} + \beta_1(x_{i1}/\sqrt{h_i}) + \beta_2(x_{i2}/\sqrt{h_i}) + \ldots + \beta_k(x_{ik}/\sqrt{h_i}) + (u_i/\sqrt{h_i})$$

  - FGLS中h_i是估计的，与$h(\boldsymbol{x})$函数形式有关，但函数可能是不对的

对GLS和FGLS估计的认识

- GLS估计具有一致性和无偏性

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + u,$$

$$y_i/\sqrt{h_i} = \beta_0/\sqrt{h_i} + \beta_1(x_{i1}/\sqrt{h_i}) + \beta_2(x_{i2}/\sqrt{h_i}) + \ldots + \beta_k(x_{ik}/\sqrt{h_i}) + (u_i/\sqrt{h_i})$$

  - 零条件均值假定下$\boldsymbol{x}$的任意恒为正的函数$h(\boldsymbol{x})$,均与误差项不相关，加权误差与加权元无关，GLS估计量一致和无偏
- FGLS估计是有偏的:因$\hat{h}_i$是估计的且$h(\boldsymbol{x})$的假定形式可能错误
  - FGLS估计依赖于估计的$\hat{h}_i$
  - 估计的$\hat{h}_i$是$\boldsymbol{x}$和y的非线性函数，FGLS估计通常有偏

$$\log(u^2) = \alpha_0 + \delta_1 x_1 + \delta_2 x_2 + \ldots + \delta_k x_k + e, \rightarrow \hat{g}. \rightarrow \hat{h}_i = \exp(\hat{g}_i).$$

OLS估计还是FGLS估计

- 通常不从估计量方差角度评价OLS和FGLS估计的有效性(有争议)，因估计的$h(x)$可能错误

$$y_i/\sqrt{h_i} = \beta_0/\sqrt{h_i} + \beta_1(x_{i1}/\sqrt{h_i}) + \beta_2(x_{i2}/\sqrt{h_i}) + \dots + \beta_k(x_{ik}/\sqrt{h_i}) + (u_i/\sqrt{h_i})$$

- 在异方差极为严重时即使方差函数$h(x)$错误，FGLS估计更好
  - 通常，稳健的FGLS标准误远小于稳健的OLS标准误。FGLS更有效

- 两个估计的正负符号一致，倾向选择FGLS估计
- 两个估计的正负符号不一致且均显著，一般表明零条件均值假定不满足，原始模型可能设定错误

利用BootStrap估计回归标准误

- 利用bootstrap估计标准误(自举标准误)
  - 以下步骤重复m次
    - 得到一个自举样本：对样本(容量n)有放回地随机抽取n个观测
    - 得到参数估计 $\hat{\theta}^{(i)}$
  - 得到： $\{\hat{\theta}^{(i)}: i=1,2,\ldots,m\}$
  - 自举标准误： $$se(\hat{\theta})=\sqrt{\frac{1}{m}\sum_{i=1}^{m}(\hat{\theta}^{(i)}-\bar{\hat{\theta}})^2}$$

例：教育回报率回归系数的bootstrap估计

```
> lgWAGE_lm1<-lm(log(WAGE)~EDUC+EXPER,data=Data)
> (usualEst<-lgWAGE_lm1$coefficients)
(Intercept)        EDUC       EXPER 
 0.21685438  0.09793557  0.01034695 
> #does a case bootstrap resampling for regression models
> library("boot")
> set.seed(123456)
> lgWAGE.boot<-Boot(lgWAGE_lm1,R=999)
> head(lgWAGE.boot$t)
     (Intercept)       EDUC       EXPER
[1,]   0.3211129 0.08983931 0.010105090
[2,]   0.4875154 0.07935380 0.009400090
[3,]   0.3184197 0.09051269 0.009647856
[4,]   0.1242433 0.10446328 0.008900771
[5,]   0.2590597 0.09493494 0.009926532
[6,]   0.1109333 0.10909502 0.009104380
> hist(lgWAGE.boot$t[,2],freq=FALSE,main="教育回报率估计的分布") #教育回报率的估计值
> lines(density(lgWAGE.boot$t[,2]))
> colMeans(lgWAGE.boot$t)
(Intercept)        EDUC       EXPER 
 0.21650307  0.09800103  0.01042714 
> bootCI<-apply(lgWAGE.boot$t,2,FUN=function(x) quantile(x,c(0.025,0.975)))
> print(bootCI,digits=3)
      (Intercept)   EDUC   EXPER
2.5%      -0.0276 0.0825 0.00744
97.5%      0.4387 0.1145 0.01351
```

```
> lgWAGE.boot

ORDINARY NONPARAMETRIC BOOTSTRAP


Call:
boot::boot(data = dd, statistic = boot.f, R = R, .fn = f, parallel = parallel_env,
    ncpus = ncores)


Bootstrap Statistics :
      original        bias    std. error
t1* 0.21685438 -3.513131e-04 0.118514845
t2* 0.09793557  6.545463e-05 0.008381355
t3* 0.01034695  8.019676e-05 0.001622759
```